# Investigating the Geographical Generalizability of AI-based Wildfire Spread Prediction Models

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**Abstract**

Wildfire prediction models are critical for risk management and emergency responses, yet their geographical generalizability remains poorly understood. This study investigates the ability of U-Net, a convolutional neural network-based model, to predict wildfire spread across different regions in the contiguous United States using geographically stratified samples. We extracted 64km x 64km samples from four distinct regions (Western US, Southern US, North Central US, and Northeast US), all spanning 2012-2020, following the pipeline used to create the state-of-the-art Next Day Wildfire Dataset. Through spatial cross-validation experiments, we trained U-Net on data from three regions while testing on the excluded fourth region. To confirm our results, we evaluated a model trained on the entire Next Day Wildfire Dataset on all regions independently and revealed significant geographical bias in model performance: U-Net achieved high accuracy in fire-prone Western US regions but exhibited poor generalization to regions with little representation in training data. When trained on geographically imbalanced datasets, the model failed to predict any fire pixels accurately in the Northeast US, while maintaining near-baseline performance in the Western US; however, when trained on a geographically balanced dataset, the model predicted accurately in the Northeast US. These findings highlight the limitations of current wildfire prediction approaches. Models trained on datasets with significant geographical imbalance perpetuate regional biases that hinder their applicability across the contiguous US, demonstrating the need to account for spatial heterogeneity in datasets either by creating more geographically balanced datasets or new training paradigms to ensure reliable prediction in less fire-prone regions.

**Introduction**

The 2020 California wildfires have sparked public concern over increasing wildfire risks in the United States. Over the last few decades, wildfires in the Western United States have increased in frequency, duration, and length of season (Westerling et al., 2006; Cunningham et al., 2024). The primary reason for this increase has been attributed to human-caused climate change, as studies have found that anthropogenic warming has contributed twice as much to fire weather than natural variability in climate (Zhuang et al., 2021). The costs of such wildfires are extreme. Economically, fire-fighting expenditures in the US reach $1 billion annually (Whitlock, 2004). Environmentally, wildfire carbon dioxide emissions have increased by 60% across the world from 2001 to 2023 (Jones et al., 2024). Predicting wildfire spread, or using geospatial machine learning (ML) models to predict where wildfires will spread the next day, is crucial for supporting faster emergency responses as well as for informing targeted measures for fire-prone areas (Tymstra et al., 2020).

There have been many different approaches to wildfire spread prediction over the past decades. Early efforts, including the use of traditional statistical and ML methods like logistic regression (Moghim & Mehrabi, 2024) and decision trees (Singh et al., 2023) struggle with the dynamic nature of environmental conditions. Physics-based models, such as (Vinodkumar et al., 2021) offer more detailed outputs, but require large amounts of computational power and environmental data. Recently, deep learning approaches have emerged for wildfire spread prediction, such as CNNs (Li et al., 2023) and other variants (Radke et al., 2019; Marjani et al., 2024).

One challenge in training models for geospatial tasks is capturing spatial heterogeneity, a property that refers to the uneven distribution of characteristics across geographic regions (Goodchild et al., 2021), while maintaining its replicability—the ability to generalize and achieve consistent results across different geographic regions (Li et al., 2024). Despite the fact that both achieving model replicability and capturing spatial heterogeneity continues to pose a problem for geospatial models, the topic has been neglected in literature about wildfire spread prediction (Konkol et al., 2019). Being able to accurately predict fire-prone areas in all geographic regions is particularly crucial due to the imbalanced nature of wildfire datasets which

stems from certain geographic regions experiencing more wildfires than others. Thus for wildfire prediction, the ability of models to generalize needs to be evaluated. For example, the Next Day Wildfire Dataset is a large-scale, state-of-the-art dataset of historical wildfires. The dataset (Huot et al., 2022) aggregates almost a decade's worth of remote-sensing data across the contiguous United States. However, the dataset is not balanced in terms of samples per region, leading to a skewed distribution of data towards the Western US. Therefore, this study aims to investigate the effects of geographically imbalanced data on the generalizability of U-Net in predicting wildfire spread. We will compare the performance of U-Net when trained on and tested in different regions in the contiguous US to evaluate how well it generalizes to new regions.

**Statement of Purpose**

The objective of this study is to evaluate the effect of geographical imbalance in a training dataset on the ability of a U-Net model to generalize wildfire spread prediction across different regions in the contiguous US.

**Role of Student and Mentor**

My mentor gave me the idea to test wildfire spread prediction models; I decided on testing the models' geographical generalizability. My mentor suggested performing a procedure involving spatial cross validation. I was the one who decided on training models on three regions and training on the third. I decided on the geographical locations of data collection, models to test, and other ablation studies (like testing models trained on Next Day Wildfire on individual regions). It was my idea to create a geographically balanced dataset and evaluate it in comparison to the Next Day Wildfire dataset. The actual implementation of the code was done by me. I gathered and recorded the data independently, and I analyzed the data independently. I formulated my conclusion about the need for better paradigms for geographical generalizability independently. My mentor guided me through the research process, checking in weekly and proofreading all work.

**Materials and Methods**

Following Huot et al.'s method for extracting samples (Huot et al., 2022), we used Google Earth Engine (Gorelick et al., 2017) to fetch data from four different locations: the Western US, the Southern US, the North Central US, and the Northeastern US. We created bounding boxes for each region, with coordinates stretching from (30, -124) to (49, -104), (25, -102) to (37, -76), (43, -104) to (48, -87), and (37, -82) to (45, -71), respectively (Figure 1). These bounding boxes were determined by rough geographic location due to the differences in fire characteristics between them. The 64km by 64km samples are extracted from January 1, 2012 to January 1, 2020, and each sample is randomly cropped to 32km by 32km for diversity in fire locations within each sample.

Each sample contains ten features: elevation, wind direction, wind speed, minimum temperature, maximum temperature, humidity, precipitation, vegetation, population density, and the previous fire mask. The previous fire mask consists of binary values indicating the wildfire's location from the previous day. The data was compiled from various remote sensing imagery datasets from the data catalog on Google Earth Engine. We selected the data sources to match those of Huot et al. for consistency:

- Elevation data are from the Shuttle Radar Topography Mission (SRTM) (Farr et al., 2007).
- Weather data are from the University of Idaho Gridded Surface Meteorological Dataset (GRIDMET) (Abatzoglou, 2012).
- Vegetation data are from the Suomi National Polar-Orbiting Partnership (S-NPP) NASA VIIRS Vegetation Indices (VNP13A1) dataset (Vermote, 2023).
- Population data are from the Gridded Population of World Version 4 (GPWv4) data set by the Center for International Earth Science Information Network (CIESIN) (Center for International Earth Science Information Network, 2017).
- Historical fire data are from the MOD14A1 V6 data set (Giglio & Justice, 2021).

Drought data was excluded as it was frequently causing batches of data to contain missing samples.

The time span of data collection was kept the same as that of the Next Day Wildfire Dataset for consistency and for compatibility with the satellite data from which the samples were collected. The distribution of data can be seen in Figure 6a and Figure 6b. There were significantly more samples in the Southern US region than the other regions, seeing as the Southern US made up 57.7% of the total number of samples collected. Even though the Southern US was the most fire-prone, the Western US had the highest average fire pixels per sample at 39.26, reflecting the amount of large wildfires in both regions.

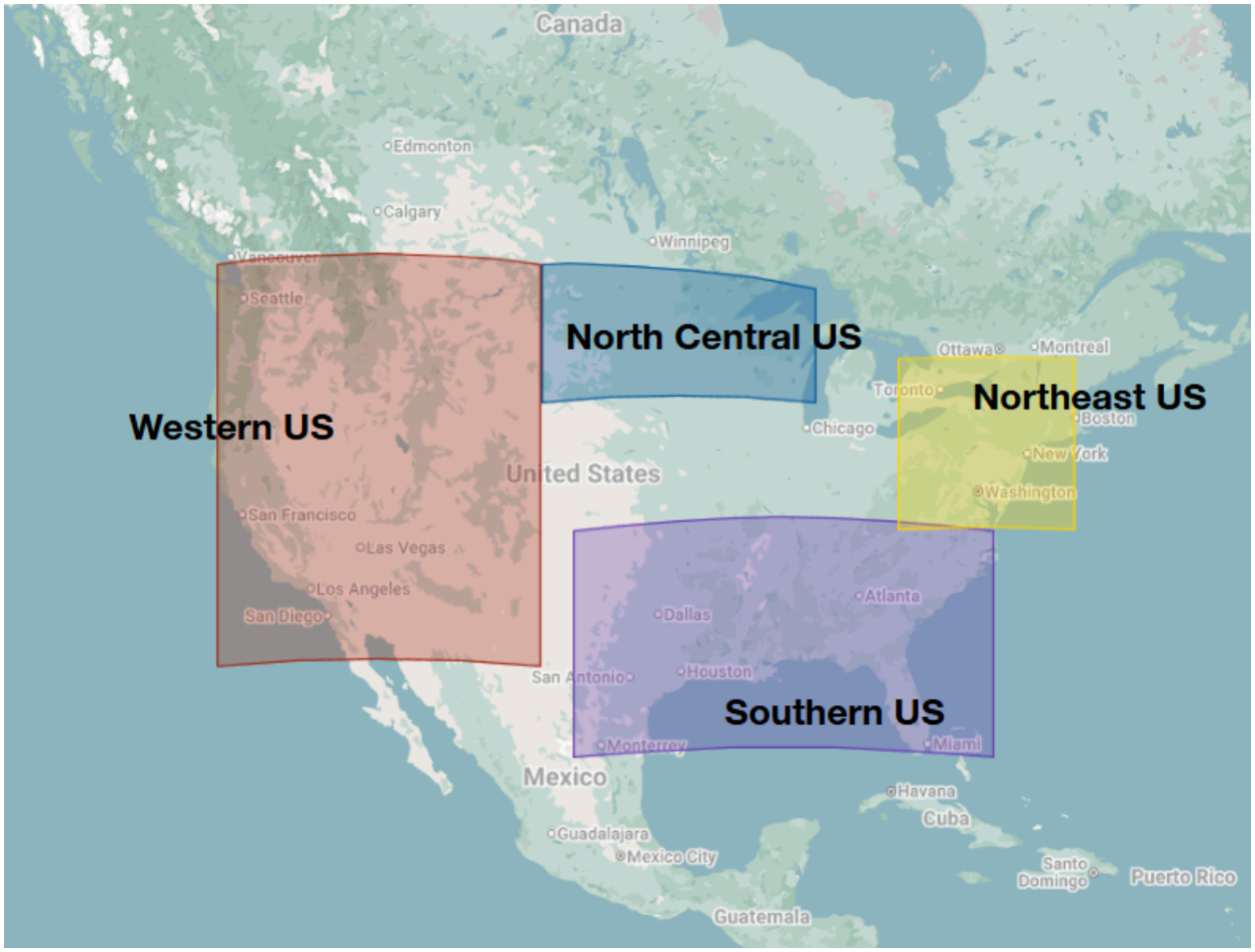


Figure 1: Visualization of bounding boxes, plotted in Google MyMaps Editor.

The wildfire spread prediction task is presented as an image segmentation task wherein the model, using the features previously mentioned, outputs a binary mask consisting of 1s or 0s (fire/no fire pixels). U-Net is an encoder-decoder network used primarily for image analysis tasks, especially image segmentation (Ronneberger et al., 2015). U-Net's architecture features

an expanding and contracting path, each connected through skip connections. These skip connections allow U-Net to preserve the spatial information lost during the downsampling process, which makes localization in image segmentation tasks more precise. The U-Net model was created using Pytorch following Fitzgerald et al.'s implementation (Fitzgerald et al., 2023).

Each U-Net model was trained for 20 epochs. We used a sigmoid focal loss function due to the severe class imbalance. The data was split by randomly separating the weeks in our time frame in an 8:1:1 ratio between training, evaluation, and testing sets. Thresholds for binary classification were determined by evaluating the F1 scores in intervals of 0.05 and selecting the threshold that yielded the best F1 score of the 20 epochs.

Besides the fire masks, the values of each feature are clipped between a minimum and maximum clip value to remove extreme, physically impossible values. The clipping values are based on either physical knowledge of the 0.1% and 99.9% percentile, depending on the feature. Then, each feature is normalized following z-score normalization (i.e., subtracting the mean and dividing by standard deviation).

To test the generalizability of the model, we conducted a spatial cross-validation study consisting of two parts. Firstly, the model is trained on data from all regions except one; the model is then tested on the excluded region, evaluating whether the model is able to predict accurately in unseen locations. We repeat this process so each region is excluded once.

Secondly, we trained the model on the entire Next Day Wildfire dataset, representative of all regions, and tested the model in all four regions individually; compared to the previous part of the study, we evaluated how the model performs in locations it has been trained on before. This approach also simulates how a model trained on state-of-the-art contiguous US datasets performs in different regions.

In addition, we train the model on a geographically balanced dataset and then test the model on each region individually to evaluate the effects of geographical imbalance on model performance, effectively creating a control with geographic balance. We create the

geographically balanced dataset by adding an equal amount of training samples per region into one dataset. The amount of training samples added per region was the amount of samples in the least represented region, which was the Northeast; thus, 3,141 training samples were added per region. The geographically balanced dataset had 12,564 samples compared to the 14,836 training samples in the Next Day Wildfire Spread dataset.

**Results**

Table 2: Performance of U-Net when tested on samples from the region excluded from the training data. Figure created by Student Researcher in Google Sheets.

| Training Data Region | Precision | Recall | F1 |
|---|---|---|---|
| Excluding South US | 2.54% | 4.40% | 3.22% |
| Excluding West US | 27.70% | 43.66% | 33.90% |
| Excluding Northeast US | 0.51% | 0.15% | 0.24% |
| Excluding North Central US | 4.48% | 1.32% | 2.04% |

Table 3: Performance of U-Net when tested on samples from the same region they were trained on. Figure created by Student Researcher in Google Sheets.

| Training Data Region | Precision | Recall | F1 |
|---|---|---|---|
| Excluding South US | 22.24% | 62.74% | 32.83% |
| Excluding West US | 12.77% | 11.69% | 12.20% |
| Excluding Northeast US | 20.48% | 36.80% | 26.32% |
| Excluding North Central US | 25.80% | 26.96% | 26.37% |

Table 4: Performance of U-Net on selected data regions when trained on the entire Next Day Wildfire Dataset. Figure created by Student Researcher in Google Sheets.

| Tested Data Region | Precision | Recall | F1 |
|---|---|---|---|
| Southern US | 11.37% | 3.85% | 5.75% |
| Western US | 23.89% | 47.94% | 31.89% |
| Northeastern US | 0 | 0 | 0 |
| North Central US | 3.31% | 0.52% | 0.91% |
| Combined | 24.07% | 23.83% | 23.95% |

Table 5: Results of model trained on geographically-balanced dataset and tested on different geographical regions. Figure created by Student Researcher in Google Sheets.

| Tested Data Region | Precision | Recall | F1 |
|---|---|---|---|
| Southern US | 8.90% | 4.82% | 6.25% |
| Western US | 21.88% | 45.84% | 29.62% |
| Northeastern US | 26.79% | 7.38% | 11.57% |
| North Central US | 1.70% | 0.47% | 0.74% |
| Combined | 21.19% | 24.91% | 22.90% |

Table 6a: Distribution of Training Samples and Average Fire Pixels for each region of data. Figure created by Student Researcher in Google Sheets.

| Data Region | Total Samples | Total Fire Pixels | Avg Fire Pixels per Sample |
|---|---|---|---|
| Western US | 52216 | 2050226 | 39.26 |
| Southern US | 82534 | 886261 | 10.74 |
| Northeast US | 3141 | 30073 | 9.57 |
| North Central US | 5166 | 86441 | 16.73 |

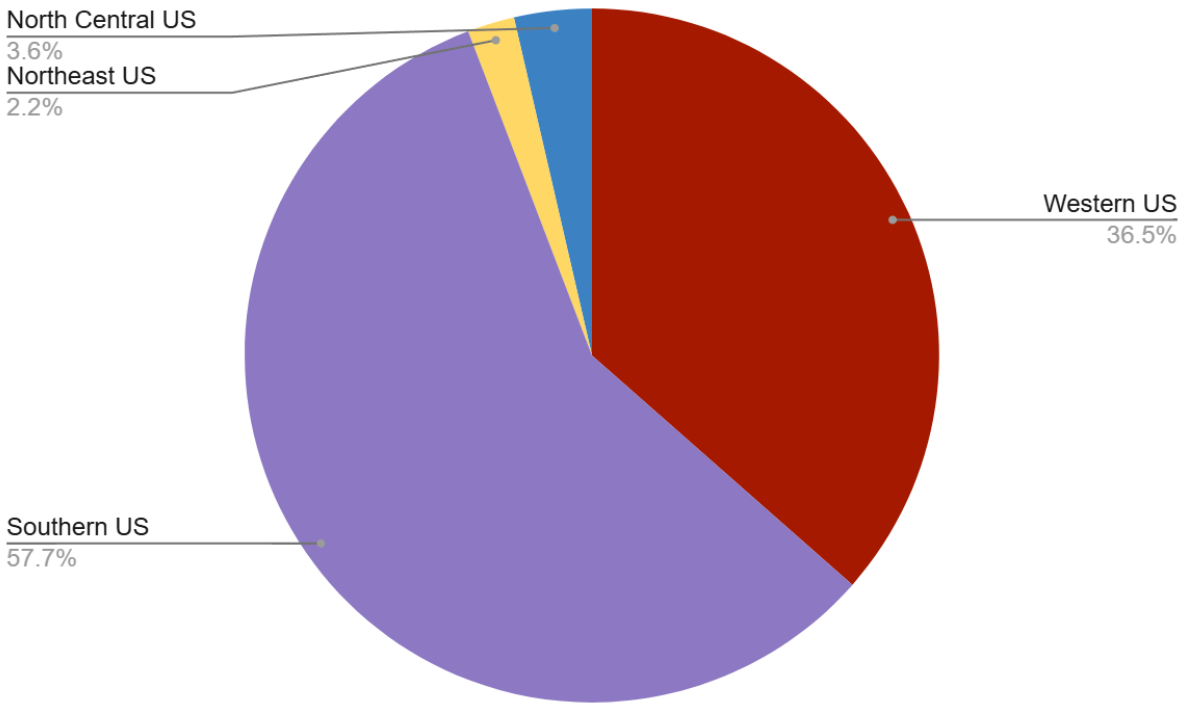


Figure 6b: Original distribution of samples per data region. Figure created by Student Researcher in Google Sheets.

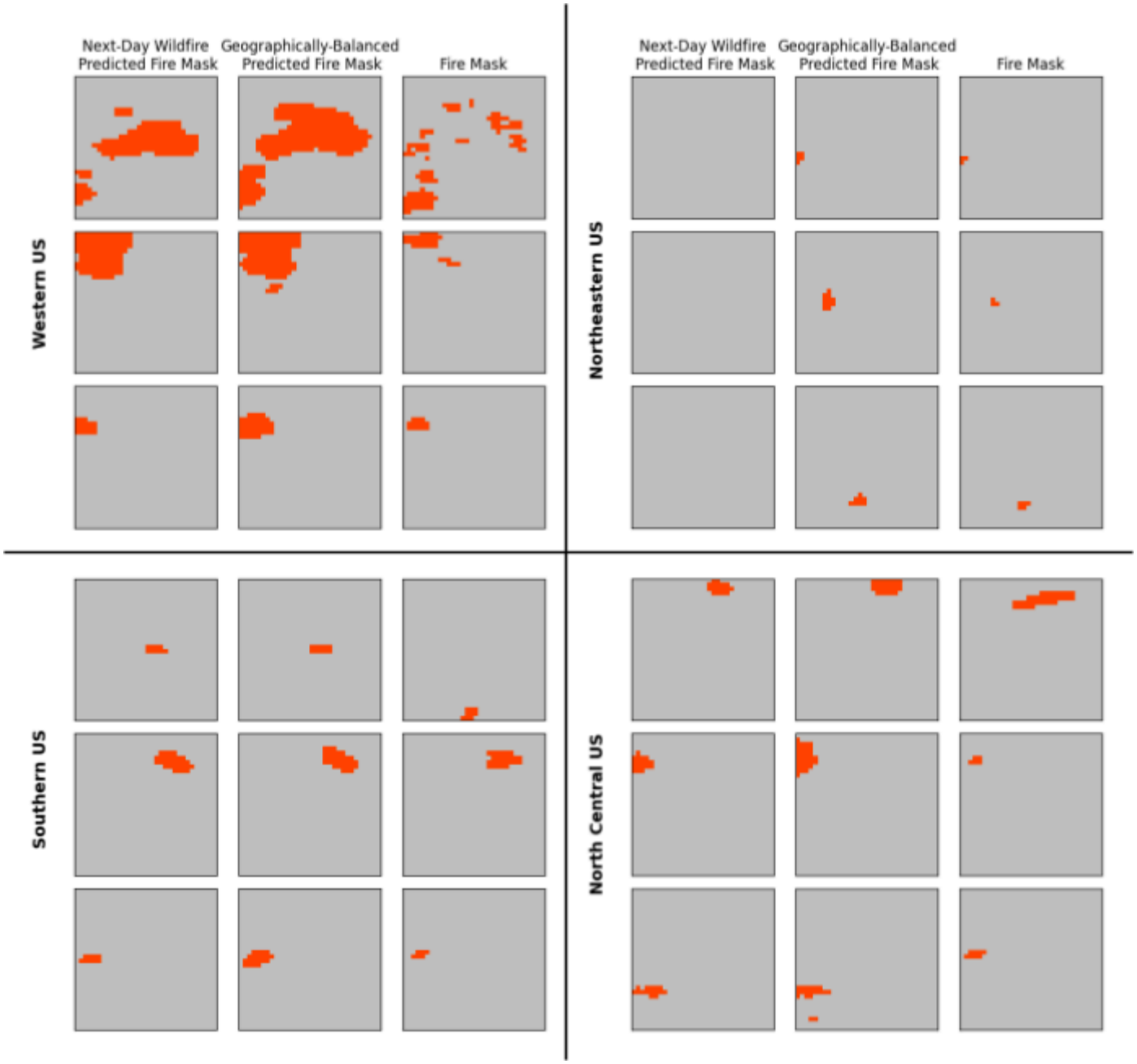


Figure 7: Examples of inferences of U-Nets trained on Next Day Wildfire Spread dataset and geographically-balanced dataset and tested on individual geographic regions. Figure created by Student Researcher in matplotlib.

**Discussion**

When the model was trained on data excluding Southern US data, Northeast US data, and North Central US data, the model performed poorly when tested on data from the excluded region, yet performed relatively well when tested on data from regions in which it was trained. This finding was consistent with our hypothesis that U-Net would have significantly better performance when testing on data from regions it has seen before.

However, when the model was trained on data excluding the Western US, the model performed poorly on its own region yet was able to predict accurately on data from the Western US despite no prior training on that region, with F1 scores of 12.20% and 31.89%, respectively. This was a surprising finding and could potentially be attributed to the characteristics of the wildfires themselves.

We further support our finding that U-Net is unable to generalize to new regions when trained on a dataset with geographical imbalance by training U-Net on the entire Next Day Wildfire Dataset and then testing the model on the entire dataset and the individual regions themselves. That the model was unable to predict any pixels in the Northeast US region accurately, yet was able to predict almost as accurately in the Western US region than it was on the entire original dataset, shows an inability to predict in regions that are not historically fire-prone and therefore not equally represented in the dataset.

The model trained on the geographically balanced dataset, in contrast to the model trained on the Next Day Wildfire Dataset, was able to predict in the Northeast US with an F1 score of 11.57%. There was also a slight increase in F1 score in the Southern US, from 5.75% to 6.25%, as well as a slight decrease in F1 scores in the Western US, 31.89% to 29.62%, and the North Central US, 0.91% to 0.74%. However, these decreases are not particularly significant, as can be seen in the visualizations of the model inferences (Figure 7). Predictions in the Western US, Southern US, and North Central US appear mostly the same, yet the difference in the Northeastern US is apparent.

**Conclusion**

Through spatial cross-validation, we found that U-Net models trained on geographically imbalanced wildfire datasets exhibit significant regional bias, performing well in fire-prone areas like the Western US while failing to generalize to regions with limited wildfire representation such as the Northeastern US. The spatial cross-validation experiments reveal that current state-of-the-art datasets like the Next Day Wildfire Dataset, despite their wide-scale coverage and feature richness, may perpetuate geographical disparities that make models unreliable to predict wildfire spread across diverse geographic contexts. However, just as training the model on the geographically balanced dataset was not enough to completely balance out the F1 scores between regions, dataset proportionality alone does not eliminate geographical disparities, and there are other factors at play that contribute to regional bias.

**Significance**

While the Next Day Wildfire Dataset is useful for training models to predict wildfire spread in fire-prone regions, the accuracy in said regions comes with the drawback of low to no accuracy in regions with little representation in the dataset. Current approaches may inadvertently perpetuate regional biases that limit model applicability across diverse geographic contexts where wildfire prediction is just as important, especially as the distribution of wildfires in the contiguous US shifts towards regions not historically fire-prone (Donovan et al., 2023). Future research should prioritize the development of more geographically balanced datasets and innovative training paradigms that can address spatial representation disparities.

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